Chapter 7 — Swaps (Part 4)

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Structure of Part 4

This section continues the advanced applications and practical dimensions of swap markets.

Topics covered:

1. Practical Hedging Insights
2. Dynamic Swap Hedging
3. Yield Curve Risk
4. Convexity and Key Rate Duration
5. Swap Spreads
6. Financial Crisis and Swap Markets
7. Modern Developments in Swaps
8. Comprehensive Formula Sheet
9. Final Chapter Intuition

This part connects theoretical swap pricing with:

- Real trading desks
- Risk-management systems
- Banking practice
- Financial crises
- Modern OTC derivatives regulation

7.11 Practical Hedging Insights

In practice, swaps are rarely used alone.

Financial institutions combine swaps with:

- Futures
- Forward Rate Agreements (FRAs)
- Treasury bonds
- Eurodollar futures
- Interest rate options
- Currency forwards
- Swaptions

This creates sophisticated hedging structures.

Dynamic Nature of Swap Risk

A swap position is not static.

Its value changes continuously because:

- Interest rates move
- Yield curves change shape
- Credit spreads fluctuate
- Volatility changes
- Exchange rates move

Thus swap portfolios require:

Continuous Risk Management

Dynamic Hedging

Swap dealers use:

Dynamic Hedging

Meaning:

- Hedge positions are adjusted continuously through time.

For example:

- If interest rates change,
- Swap duration changes,
- Hedge ratios must be recalculated.

Example — Dynamic Hedge Adjustment

Suppose:

- Bank receives fixed in swap
- Initially hedged using Treasury futures

If rates fall sharply:

- Swap duration increases
- Existing hedge becomes insufficient

Thus:

- Additional futures contracts may be required.

Why Dynamic Hedging Is Necessary

Swap risk depends on:

- Time remaining to maturity
- Yield curve shape
- Interest-rate volatility
- Convexity

Therefore:

A hedge that works today may fail tomorrow.

Yield Curve Risk

One of the biggest challenges in swap hedging is:

Yield Curve Risk

Most simple hedges assume:

Parallel shifts in the yield curve

Meaning:

- All interest rates move equally.

In reality:

- Different maturities move differently.

Example — Nonparallel Shift

Suppose:

- Short-term rates rise by 2%
- Long-term rates rise by only 0.2%

Then:

- A duration hedge may perform poorly.

Because:

Different points on the curve moved differently.

Key Rate Duration

To handle yield curve risk, institutions use:

Key Rate Duration

Key rate duration measures sensitivity to:

- Specific maturities on the yield curve

Examples:

- 2-year rate sensitivity
- 5-year rate sensitivity
- 10-year rate sensitivity
- 30-year rate sensitivity

Why Key Rate Duration Is Important

A portfolio may have:

- Small overall duration exposure

but:

- Large exposure to one maturity segment.

Key rate durations provide:

More precise interest-rate risk measurement.

Convexity in Swap Portfolios

Duration measures:

First-order sensitivity

But large interest-rate changes create:

Convexity Effects

Convexity measures:

- Curvature in price-yield relationship.

Importance of Convexity

Suppose rates move sharply.

Then:

- Linear duration approximations become inaccurate.

Swap portfolios with large convexity exposure may:

- Gain more than expected
- Lose more than expected

Managing Convexity

Large dealers often hedge:

- Duration
- Convexity
- Key rate exposure

simultaneously.

This requires:

- Complex mathematical models
- Sophisticated trading systems

Swap Spread

One of the most important concepts in fixed-income markets is:

Swap Spread

Defined as:

$$\text{Swap Spread} = \text{Swap Rate} - \text{Treasury Yield}$$

for the same maturity.

Example — Swap Spread

Suppose:

- 5-year swap rate = 4.20%
- 5-year Treasury yield = 3.85%

Then:

$$4.20 - 3.85 = 0.35\%$$

Thus:

Swap spread = 35 basis points

Interpretation of Swap Spread

Swap spreads reflect:

- Credit risk
- Banking system conditions
- Liquidity conditions
- Supply and demand
- Interbank funding stress

Why Swap Rates Exceed Treasury Rates

Typically:

$$\text{Swap Rate} > \text{Treasury Yield}$$

because:

- Treasury securities are nearly risk-free
- Swap markets involve interbank credit exposure

Thus swap spreads compensate for:

Credit and liquidity risk.

Financial Crisis and Swap Markets

Swap markets became critically important during the:

2008 Global Financial Crisis

The crisis revealed:

- Massive interconnectedness among financial institutions

- Large OTC derivatives exposures
- Systemic counterparty risk

AIG Example

One famous example:

AIG (American International Group)

AIG sold enormous amounts of:

Credit Default Swaps (CDS)

When mortgage markets collapsed:

- AIG faced huge collateral calls
- Systemic financial risk increased dramatically

Government intervention became necessary.

Lessons from the Financial Crisis

The crisis highlighted several weaknesses:

1. Excessive Leverage

Financial institutions held massive derivative exposures.

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2. Lack of Transparency

OTC positions were difficult to observe.

3. Counterparty Risk

Default of one institution threatened others.

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4. Insufficient Collateral

Many exposures were undercollateralized.

Regulatory Changes After 2008

After the crisis:

Swap markets became far more regulated.

Major reforms included:

- Central clearing mandates
- Higher capital requirements
- Mandatory reporting
- Margin requirements
- Trade repositories

Central Counterparties (CCPs)

Modern swap markets increasingly use:

Central Counterparties

CCPs:

- Stand between buyers and sellers
- Guarantee contract performance
- Reduce bilateral credit exposure

Advantages of CCPs

Reduced Counterparty Risk

CCP guarantees performance.

Greater Transparency

Positions become more observable.

Better Netting

Exposures across participants can offset.

Lower Systemic Risk

Risk becomes centralized and monitored.

Disadvantages of CCPs

However CCPs also create concerns.

Concentration Risk

Risk becomes concentrated in clearinghouse.

Operational Complexity

Clearing infrastructure is expensive.

Reduced Customization

Standardization may limit flexibility.

Modern Benchmark Transition

Historically, most swaps used:

LIBOR

However LIBOR faced:

- Manipulation scandals
- Declining interbank lending activity

Thus markets transitioned toward:

SOFR (Secured Overnight Financing Rate)

in the United States.

SOFR vs LIBOR

Feature	LIBOR	SOFR
Credit Risk	Includes bank credit risk	Nearly risk-free
Underlying Market	Interbank lending	Repo market
Tenor Structure	Multiple maturities	Overnight rate

Implications of Benchmark Transition

Transitioning from LIBOR to SOFR affected:

- Swap pricing
- Hedging models
- Valuation systems
- Risk management frameworks
- Contract documentation

This became one of the largest changes in derivatives markets in decades.

Swap Market Size and Importance

The global swap market is enormous.

Swaps influence:

- Bond pricing
- Corporate financing
- Monetary policy transmission
- Banking system stability
- Global capital flows

Thus swaps are foundational to:

Modern Financial Architecture

Relationship Between Swaps and Monetary Policy

Central bank policy strongly affects swap markets.

Examples:

- Federal Reserve rate hikes
- Yield curve changes
- Liquidity injections
- Quantitative easing

All directly influence:

- Swap rates
- Swap spreads
- Hedging demand

Practical Trading Insight

Professional swap traders monitor:

- Yield curves
- Central bank policy
- Inflation expectations
- Funding markets
- Credit spreads
- Treasury markets
- Futures markets

because all these affect swap valuation.

Integrated Nature of Derivatives Markets

Swap markets are closely linked with:

- Treasury markets
- Repo markets
- Futures markets
- Options markets
- FX markets
- Credit markets

Thus understanding swaps requires:

Understanding the broader financial system.

Comprehensive Formula Sheet

Interest Rate Swap Value

Fixed Payer

$$V_{swap} = B_{floating} - B_{fixed}$$

Floating Payer

$$V_{swap} = B_{fixed} - B_{floating}$$

Fixed Bond Value

$$B_{fixed} = \sum_{i=1}^{n} Ke^{-r_i t_i} + Le^{-r_n t_n}$$

Currency Swap Value

$$V_{swap} = B_{domestic} - S_0 B_{foreign}$$

Swap Spread

$$\text{Swap Spread} = \text{Swap Rate} - \text{Treasury Yield}$$

Key Exam Concepts from Part 4

You must understand:

- Dynamic hedging
- Yield curve risk
- Key rate duration
- Convexity
- Swap spread
- CCPs
- Financial crisis implications
- LIBOR vs SOFR
- Systemic risk in swaps
- OTC derivatives regulation

Most Important Intuitions

1. Swap Risk Is Dynamic

Swap exposure changes continuously with markets.

2. Yield Curves Matter More Than Single Rates

Different maturities move differently.

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3. Swap Markets Reflect Financial System Health

Swap spreads and funding markets reveal stress.

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4. Credit Risk Is Central in OTC Markets

Counterparty exposure can become systemic.

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5. Modern Swap Markets Are Highly Regulated

Post-2008 reforms transformed derivatives markets.

Final Intuition of Chapter 7

This chapter develops one of the most important frameworks in derivatives markets.

Core ideas:

1. Swaps exchange future cash flows.
2. Interest rate swaps transform interest-rate exposure.
3. Currency swaps transform currency exposure.
4. Swaps can be priced using bond valuation techniques.
5. OTC derivatives create counterparty credit risk.
6. Dynamic hedging is essential in swap management.
7. Yield curve risk is more complex than simple duration risk.
8. Swap spreads contain information about financial conditions.
9. Modern financial institutions rely heavily on swaps.
10. Swaps are central to the global financial system.